

**Design of an AI-Driven Malaria Screening and Diagnosis Support System.**

**By**

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## **CERTIFICATION**

This is to certify that ESEKHAIGBE VICTOR-CARLTON OSEREMEN with 22CD009352 has carried out this research project in partial fulfillment of the requirement of the award of Bachelor of Science Degree in Computer Science in the College of pure and Applied Sciences, Landmark University and was thoroughly supervised by my project supervisor Dr. Emmanuel Igbekele.

## **DECLARATION**

I, ESEKHAIGBE VICTOR-CARLTON OSEREMEN hereby declare that the research project submitted for evaluation for the award of Bachelor of Science (Bsc.) degree in COMPUTER SCIENCE was carried out by me.

## **DEDICATION**

This work is dedicated to God Almighty for the gift of knowledge and perseverance, and to my family for their endless support, sacrifices, and belief in me throughout this academic pursuit.

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## **CHAPTER ONE**

### **1.0 Introduction**

Malaria is among the most widespread infectious diseases among the populations of the tropical and subtropical areas of the globe and in developing countries. The disease is still posing serious health challenges because it has a high rate of transmission and also because of the long lasting complications caused by the disease that can be encountered in cases where the diagnosis and treatment is delayed. The key aspect of malaria management is very much dependent on the early and accurate diagnosis of its effects so that timely treatment may be provided and the chances of death and the spread of the disease may be minimized.

The healthcare systems in most malaria-infested regions specifically in low resource settings are experiencing chronic challenges, which render ineffective diagnosis. These challenges include inadequate medical facilities, absence of diagnostic centers and unskilled medical staff. This may mean that the conventional forms of diagnostic procedures may fail to provide timely and reliable diagnostic information leading to misdiagnosis and delayed treatment. Such limitations underscore the value of supportive diagnosing techniques that are capable of fostering clinical decision-making and improved patient outcomes.

The recent Information Technology (IT) breakthroughs, particularly, the areas of Artificial Intelligence (AI), and Machine Learning (ML), offer new opportunities to improve the

healthcare delivery process. The AIs have proven to assist the health care professionals in screening and diagnosing diseases by analyzing clinical data and making intelligent decisions (Abdar et al., 2021). The application of machine learning has also shown some promise in improving the accuracy and efficiency of diagnostics related to infectious diseases as an additional measure to traditional diagnostic tools (Ang & Bansal, 2021).

In the proposed study, attention is devoted to the design of the malaria screening and diagnosis support system as the AI-powered IT-based product to overcome the limitations issues associated with the conventional malaria diagnosis. The choice to do so is guided by the fact that there is more urgency to integrate smart systems in the healthcare industry, particularly in regions that lack diagnostic centers and human resources. The proposed system will assist healthcare professionals in making prompt and correct decisions regarding diagnosis, which will help reduce malaria and healthcare delivery through the usage of AI technologies.

## **1.1 Background of the Study**

Malaria is the infectious disease that is a parasite and has a species of Plasmodium and is spread by the bite of the infected female Anopheles mosquitoes. Throughout its decades of existence, it has remained one of the key public health issues, especially in the tropical and subtropical areas whereby both environmental and socio-economic conditions support its propagation. Despite the remarkable achievements in the control of the malaria disease, the disease still poses a great burden to the healthcare systems of developing countries because of the problems related to prevention, diagnosis, and treatment. Among these problems, the



right and prompt diagnosis is one of the crucial issues that determine the results of the disease.

In the past, diagnosis of malaria has been mainly based on clinical evaluation and laboratory-related methods. Symptom-based clinical diagnosis like fever, headache, chills, and fatigue, have been in common use particularly in resource-constrained setting. This method is however not reliable since malaria is similar to many other febrile diseases such as typhoid fever and even the viral diseases. This consequently causes correct diagnosis and wrong treatment when only clinical signs are used, which exacerbates patient outcomes and leads to drug resistance (Bhattacharyya & Kim, 2019).

Microscopy has been considered to be the gold standard in the diagnosis of malaria. The technique entails observation of stained blood smears under a microscope to identify malaria parasites, species of parasites and the density of parasites. Although microscopy is important in offering detailed diagnostic information, it is labor intensive and time consuming, and it is very reliant on the skill and experience of the laboratory employees. The lack of trained microscopists, low quality equipment and unreliable power supply in most of the endemic areas greatly diminish the quality and accuracy of microscopy-based diagnosis (Al-Doghman & Al-Fatlawi, 2020).

Rapid Diagnostic Tests (RDTs) were introduced as an alternative form of diagnosis to overcome some of the limitations that are associated with microscopy. RDTs recognize malaria antigens in the blood and are set to give results within minutes, usually without highly complicated lab facilities. These have resulted in the RDTs gaining popularity in primary care facilities and remote locations due to their simplicity and speed of providing

an answer. However, the quality of the RDTs operations are also limited including the sensitivity limitations at low concentrations of the pathogen and the lower accuracy of the tests due to the presence of poor storage facilities and changes in the quality of the tests (Al-Doghman & Al-Fatlawi, 2020).

The reason is that the classical modes of diagnosis of malaria have repeatedly been limited thus leading to the need to explore other and supplementary modes of diagnosis that might increase the accuracy and effectiveness of diagnosis of malaria. The last several years have seen Information Technology (IT) becoming a revolution in the healthcare systems of different parts of the world. Such technologies as Artificial Intelligence (AI), Machine Learning (ML), and big data analytics make the creation of intelligent systems to help make medical decisions based on analyzing complex datasets and revealing patterns that cannot be determined by humans possible (Razzak et al., 2019).

Artificial Intelligence is a procedure in which the computer systems are able to perform tasks, which computer system otherwise is not supposed to accomplish, including learning, reasoning and problem solving. Medical imaging, disease prediction, clinical decision support, and automated diagnosis are a few of the areas that AI-based systems have been applied successfully in the healthcare sector. Such systems leverage machine learning models to train on previous data and provide predictive or classification results that can assist a medical practitioner to make effective decisions (Abdar et al., 2021).

The application of AI in the process of diagnosis of diseases became of special interest due to the fact that it is likely to enhance the accuracy, reduce the human factor, and make the process more efficient. Neural networks, decision trees, support vector machines, and deep

learning models are machine learning techniques that are showing good promising outcomes in detecting and classifying infectious diseases. Research has demonstrated that AI-powered diagnostic tools can be used to complement other diagnostic tasks by offering a consistent and objective evaluation of the data and not a subjective one (Ang & Bansal, 2021).

As applied to malaria diagnosis, AI-based solutions have been investigated in different data sets, such as blood smart photos, patient history information, and laboratory outcomes. The objectives of these approaches are to automate or assist in the diagnostic activities by detecting malaria-related trends and decision support to the healthcare workers. In spite of encouraging research results, there are still few places of AI-supplied malaria diagnostics systems implementation in clinical practice. Difficulties with unstructured systems, insufficient technical infrastructure and issues with system reliability and usability have made the wide adoption a challenge (Bhattacharyya & Kim, 2019).

Also, a significant portion of current AI-based malaria work is focused on the development of algorithms and their performance analysis, but less on how the system is designed and integrated into the health care workflow. The gap underscores the necessity of systematic frameworks that theorize the application of AI in the context of malaria screening and diagnosis as a decision support tool. These frameworks ought to focus on usability, reliability and compatibility with the current healthcare practices and especially in low resource settings that feature malaria.

With this in mind, the context of this paper defines the topicality of introducing new IT tools, in particular Artificial Intelligence, in malaria diagnostics. The study is aimed at

making a contribution to the current effort of enhancing diagnostic practice, assisting healthcare providers, and improving malaria management by developing an AI-based malaria screening and diagnosis support system. It is hoped that the proposed framework will become a cornerstone framework that may be used to develop future and improve the implementation and assessment of AI-based diagnostic systems in malaria-endemic areas.

## **1.2 Statement of the Problem**

Malaria is a significant social health issue in most developing nations, especially in tropical and sub-tropical areas whereby malaria is rife. In spite of the fact that malaria is preventable and treatable, there are still difficulties in the aspect of timely and proper diagnosis that hinder its effective management. In most of the malaria-infested world, medical institutions still use the traditional forms of diagnosis that are either slow, resource consuming, or they require highly skilled individuals. Such constraints play a major role in the delay of diagnosis, misdiagnosis and inappropriate treatment thereby leading to higher morbidity, mortality as well as further transmission of the disease.

Conventional methods of diagnosing malaria like microscopy and rapid diagnostic test (RDTs) have their significant weaknesses. Although it can determine the species of parasites and the parasite density, microscopy needs qualified laboratory staff, proper equipment, and the constant quality of the laboratory conditions, which are not always present in low-resource healthcare facilities. The disadvantages of RDTs are that they are not as sensitive as other diagnostic methods, especially at low parasite concentrations, and

storage and test quality can influence results. The limitations undermine the credibility and the general performance of the traditional methods of diagnosis in places where malaria prevalence is the most significant (Al-Doghman and Al-Fatlawi, 2020).

What complicates the diagnostic issue is the fact that malaria has common clinical features with other febrile diseases and so, symptom-based diagnosis is ineffective. This results in wrong treatment of patients by the healthcare providers and thereby causing delayed or wrong treatment to the patient. This state of affairs puts more strain on already strained healthcare systems and negates continuous malaria control measures. Research has highlighted that diagnostic error has been an imperative challenge to successful malaria control in endemic areas (Ang and Bansal, 2021).

The new developments in the field of Artificial Intelligence (AI) have shown that it has a great potential in solving tricky diagnostic issues in medical practice. DSSs based on AI have also been demonstrated to improve the accuracy of the diagnoses, increase efficiency, and assist clinicians with the help of data-driven insights obtained with the help of large data sets (Abdar et al., 2021). Although these achievements have been made, there is still not much AI use in malaria screening and diagnosis, especially in structured and flexible models that can be adapted to low-resource environments. It is this lack of contemporary IT implementation that has led to the research to investigate designing of an AI-based malaria screening and diagnostic support system that will enhance diagnostic practices and assist healthcare professionals in malaria prone settings.

### **1.3 Aim and Objectives**

The aim of this project is to create an AI-Driven Malaria Screening and Diagnosis Support System.

### **Objectives of the Study**

The objectives of this project are:

1. To review literature on AI-based screening and diagnosis system for malaria.
2. To design a model illustrating an AI-driven framework for malaria screening and diagnosis.
3. To implement the designed framework in a simulated environment to assess its functionality.
4. To evaluate the framework using criteria such as diagnostic accuracy, efficiency and reliability.

### **1.4 Significance of the Study**

This study focuses on applying innovative IT solutions to problems faced in the diagnosis of malaria. With the continued digitization of the healthcare ecosystem, there is a need for smart systems that can assist practitioners in making clinical decisions and improving the efficiency of diagnosis, particularly in low resource settings. This study aims to fill this gap by using AI as a support tool in the screening and diagnosis of malaria.

From an IT problem-solving perspective, the study illustrates how AI based systems can be applied to and through complex real-world problems that have a high degree of uncertainty, demand a high degree of accuracy, and have low availability of human

resources. The framework proposes a way that data driven techniques can be applied to modern IT solutions to improve the efficiency, consistency, and reliability of healthcare provisioning.

The study is important to healthcare practitioners because it aims at building a supportive diagnostic framework, and not a replacement, for medical professionals. The system is designed to assist clinicians with timely and regular diagnostic information and is expected to help reduce treatment delays and improve decision support, particularly in settings with limited qualified staff and diagnostic tools.

In terms of scholarship, the study adds to the body of knowledge with a designed framework that attempts to integrate artificial intelligence in malaria screening and diagnosis. The study also provides the basis for subsequent research and the construction of systems in intelligent healthcare solutions. The study mainly demonstrates the usefulness of modern information technology in solving persistent problems in public health and in the healthcare delivery system.

## **1.5 Scope of the Study**

This paper is confined to the conceptual development of AI-based malaria detection and diagnosis assistance system. This research aims at coming up with a framework that will assist healthcare professionals in the effective diagnosis of malaria by availing prompt and data-driven decision support. This system will maximize the efficiency of diagnosis, minimize the possibility of human error and improve the general decision-making processes in malaria-infested countries.

The regions of interest to the study, geographically, are the regions where malaria is common especially the tropical and subtropical areas with limited health care facilities. Although the framework is supposed to be flexible to different contexts, it mainly addresses low-resource contexts where traditional diagnostic procedures might not be effective or reliable.

The paper addresses systematically the design and simulation of the AI-based framework. This involves outlining the system design, workflow design, and testing the structure based on the performance measures that have been chosen like diagnostic accuracy, reliability and efficiency. There is no full software implementation, real-time deployment, or clinical trials covered in the study; the performance of the proposed framework is assessed on the basis of simulation and available data to prove the viability and efficiency of the suggested framework.

In general, the extent of the research provides a clear view of the project, as the conceptual design, AI integration, and decision support in malaria diagnosis are discussed, although the implementation and clinical validation are future research issues.

## **1.6 Justification of the Study**

This study is needed due to the continued difficulty of diagnosing cases of malaria, especially in the resource-limited areas where the most common diagnostic methods are slow, make mistakes, and are hard to access. This study suggests the first step in providing a framework to support the accuracy and efficiency of diagnosis by using artificial



intelligence in malaria screening and diagnosis support systems and to help healthcare workers make more accurate and timely decisions.

This study adds to the body of knowledge by providing a framework to help integrate the use of Artificial Intelligence into systems designed to support healthcare decisions. Most other previous works have focused on using Artificial Intelligence to develop algorithms for the detection of disease; however, this research focused on design, system-level design, and integration of the entire system, including the workflow, the system architecture, and the simulation of the integrated system. This type of research is intended to help additional research and the development of systems for the use of Artificial Intelligence in the diagnosis of other diseases, particularly in areas of the world where malaria is prevalent.

In addition, the research paper shows that modern IT solutions can be applied to the issues of public health and explains why AI can be used to enhance the current diagnostic techniques instead of substituting human skill. This study offers useful information to scholars, IT programmers, and healthcare policymakers who want to use intelligent systems to enhance healthcare delivery by reporting the design, consideration of implementation, and assessment of an AI-driven framework.

Finally, the research should serve the purpose of bridging the existing gap between theory and practice AI implementation and making contributions in the field of academia and the potential practical effects of the research in malaria development.

## **1.7 Limitations of the Study**

Every study has its constraints. In this case, the entire system is theoretical, so the proposed AI-driven malaria screening and diagnosis support system is conceptual, meaning the system will lack the complete construction, training, and deployment of the machine learning models. Therefore, performance evaluations are based on system literature and theory, and not actual clinical testing. In addition, the reliance on previous research and secondary sources of data may lead to biases and an inability to capture the differences of populations and various health systems and the way malaria presents across diverse regions.

Moreover, the study is limited to a few clinical parameters and does not address all potential analytes, diagnostic variables and sophisticated laboratory procedures related to malaria diagnosis. Additionally, although the proposed system is meant to serve as a decision support system and not a substitute for healthcare providers, this means that the final decision of a diagnosis rests on the clinical parameters, and not this system, the provided system is meant to assist. Nonetheless, the importance of the study remains, for the research is meant to provide a basis or a structured outline for future system development and implementation.

## **1.8 Definition of Terms**

Artificial Intelligence (AI): A system that assists in diagnostics and provides feedback by analyzing data, in this case, at the rapid and complex level, through the use of integrated reasoning and other skills parasite.

Machine Learning (ML): Subset of AI that teaches computer systems through data and enhances their capabilities over time without explicit programming. In this study, ML algorithms serve to identify and set the framework for the use of data in the context of malaria.

Malaria Screening: Identification of individuals for the presence of malaria for customers by the presence of Plasmodium parasites.

Malaria Diagnosis: Process of confirming the presence of malaria in a patient, including the identification of the parasite species, and in some rare cases, the level of infection.

Rapid Diagnostic Test (RDT): A set of tools that makes use of a set of standards for the detection of malaria in the blood sample antigens and produces results without the use of complex laboratory infrastructure.

Microscopy: is the gold standard for malaria diagnosis and has the involvement of a laboratory technique that includes making a thin stained blood smear and then examining it under the microscope to find the malaria parasites.

Decision Support System (DSS): A system based on computer technology that helps analyze data and provides actionable suggestions for the healthcare professionals for making informed decisions.

Diagnostic Accuracy: the ability of a diagnostic test to correctly classify the presence or absence of a disease to a person. Here, it relates to how well the AI system detected malaria cases.

Resource-Constrained Settings: Places, especially, in most malaria- endemic areas, where healthcare systems lack trained staff, diagnostic equipment, and other basic infrastructure.

## **CHAPTER TWO**

### **2.1 Malaria**

Malaria is a deadly infectious disease, which is incurred by protozoan parasites of genus Plasmodium and which is spread in humans due to the biting of infected female Anopheles mosquitoes. Plasmodium falciparum, Plasmodium vivax, Plasmodium

malariae, *Plasmodium ovale*, and *Plasmodium knowlesi* are the major *Plasmodium* species that cause malaria infection to man. *Plasmodium falciparum* stands out as the most virulent and causes the most severe cases of malaria and the most death due to malaria, especially in sub-Saharan Africa (Phillips et al., 2017).

The transmission of malaria takes place whereby the infected mosquito penetrates through the blood of the human body with *Plasmodium* parasites. The parasites will target the liver where they reproduce and go back to the blood stream to infect red blood cells. It is at this phase of infection that the clinical signs of malaria such as fever, chills, headache, fatigue and anemia are caused. Other severe complications of malaria include cerebral malaria, organ failure, and death, particularly of populations at risk, including children and pregnant women (White et al., 2018).

Malaria has been a significant issue of concern in the public health front irrespective of the world trying to contain the disease and eventually eradicate it. The disease is common mostly in the tropical and sub-tropical areas where breeding and transmission of mosquitoes is facilitated by climatic conditions. The poverty, poor access to health services, poor disease surveillance are also socio-economic factors that make the burden of malaria worse in developing countries. According to the literature, malaria remains one of the biggest burdens on the healthcare sector, causing high hospital readmission rates, labor force losses, and high mortality in the endemic areas (Tizifa et al., 2018).

There have been strategies over the years that have been used to minimize the spread and deaths caused by malaria such as the control of vectors, chemoprevention and awareness of the people about Malaria. Although such measures have led to the reduction in malaria

rates in certain areas, the progress has been weak and irregular. The problem of early and accurate diagnosis is one of the acute problems that deter the work on malaria control, especially in low-resource environments where both diagnostic facilities and qualified specialists are scarce (Okell et al., 2020).

Proper malaria treatment can only be achieved through proper diagnosis, which allows proper medication and will limit the misuse of antimalarial drugs. Wrong diagnosis or late diagnosis may result in severe development of diseases and further transmission at the communal level. Malaria is therefore one of the unremitting problems which need better diagnostic solutions in addition to the current prevention and treatment methods. The nature, transmission, and effects of malaria are the aspects that offer a needed background regarding the analysis of the diagnostic imaginations and the discussion of novel ways to address the issue, including the utilization of Artificial Intelligence-based screening and diagnostic applications.

## **2.2 Malaria Diagnosis**

The diagnosis of malaria can be described as the procedure of determining the presence of the Plasmodium parasites in an infected person to ascertain the presence of the infection and provide the necessary treatment. Proper diagnosis is very important in the management of cases of malaria since the clinical manifestations of malaria like fever, headache, chills, and fatigue are also similar to those of other infectious diseases such as typhoid fever and viral infections. Applying clinical presentation without laboratory tests may thus result in wrong treatment choices and adverse patient outcomes (Moody, 2020).

Malaria diagnosis is not only about the patient but has a far-reaching public health goal. Early and proper diagnosis helps to treat the disease in time, minimise the severity of the disease and prevent further spreading of the disease in the society. Malaria surveillance can also be enhanced through effective diagnosis to ensure effective data collection in the monitoring of trends of malaria, analyzing interventions, and efficient allocation of health resources. In endemic countries, the accuracy of diagnosis is especially significant in helping to avoid the unnecessary or wrong use of antimalarial medication, which is one of the causes of drug resistance development (Bousema et al., 2018).

Malaria diagnosis has a number of problems, although it is significantly important and most of the burden of the disease is experienced in the low- and middle-income countries. Low availability of diagnostic centers, lack of qualified laboratory staff and lack of proper health care facilities are major barriers to effective diagnosis. In rural and resource-limited locations, healthcare-delivering professionals might not have the tools to identify malaria infection, and instead, the diagnosis outcomes may be presumed, which may negatively affect the reliability of the diagnosis (Landier et al., 2016).

The other one is the major challenge in malaria diagnosis in the characterization of low-density parasitemia which is mostly experienced by the asymptomatic people or half immune people. Conventional diagnostic modes may not accurately detect such cases and these cases may subsequently be transmitted within the population. Furthermore, diagnostic accuracy could also be made complicated by differences in species of parasites and their stages of infection. The challenges identified above demonstrate the necessity of diagnostic methods that are sensitive, specific, scalable, and flexible to various healthcare settings (Chiodini, 2014).

The effects of wrong malaria diagnosis are immense. False-positive diagnoses may lead to unneeded use of antimalarial medication, higher costs of healthcare, and delayed medication of the underlying disease. On the other hand, false-negative diagnoses can cause the development of the disease, serious complications, and the rise in the mortality rate. On the population level, diagnostic gaps that persist despite control and elimination efforts render malaria prevention weak because they enable infections that go undiagnosed to perpetuate the process of spreading malaria (Gething et al., 2017).

In the light of these constraints, the improvement of the malaria diagnostic process with the help of novel and technologically oriented solutions is becoming a matter of interest. Recent progress in computational technologies, especially Artificial Intelligence and machine learning, creates possible opportunities to improve the accuracy of diagnostics, decrease the reliance on limited human knowledge, and facilitate the decision-making process in the clinical field. These new directions give a reason to reconsider the diagnosis of malaria and go outside the conventional methods and encourage the idea of the search of AI-based screening and diagnostic tools.

### **2.3 Conventional Malaria Diagnostic Techniques**

Conventional malaria diagnostic methodologies are procedures that have been in use in the past to diagnose the presence of Plasmodium parasites in suspected malaria patients. In most health care systems, especially in the endemic countries, these methods are the



foundation of the malaria diagnosis. Malaria control and elimination programs require the use of effective diagnostic methods that would verify infection, direct treatment, and aid in diagnosis. Light microscopy and Rapid Diagnostic Tests (RDTs) represent the most common traditional diagnostic techniques that have unique operational principles, pros, and cons (Wongsrichanalai et al., 2007).

Early and correct diagnosis of malaria by the conventional methods has a number of advantages. Early diagnosis ensures that the disease is treated in time, the risk of serious complications is minimized, and further spread of the disease is reduced. It also serves to rule out the use of antimalarial drugs where they are unnecessary in the non-malarial febrile diseases and hence the risk of drug resistance is avoided. But even though they have become common, the traditional diagnostic methods have certain challenges where their accuracy, availability, and consistency may be problematic particularly in low-resource areas (Ochola et al., 2016).

### **2.3.1 Microscopy**

Microscopy is regarded as the gold standard to diagnosis malaria and it is the analyses of stained blood smears under light microscope to identify parasites of Plasmodium. There are two common types of blood smears; thick smears, which are more sensitive to low densities of parasites, and thin smears, which can be useful in identifying the species of the parasites and the parasite load. Microscopy enables training employees to differentiate the various types of Plasmodium species and determining the intensity of infection (Tangpukdee et al., 2009).

Microscopy has a high diagnostic accuracy; this is among the significant benefits of microscopy provided that skilled laboratory technicians carry out the diagnostic process. It offers specific details of the morphology of the parasites, species of the parasites and parasitemia levels, which is valuable in handling the case. Furthermore, microscopy is comparatively inexpensive in the long run in laboratories that have adequate microscopic equipment and is not based on proprietary test kits.

Microscopy has several limitations despite these strengths. Microscopic diagnosis is very reliant on the expertise and experience of the microscopist and hence the variability in the diagnosis. The endemic areas lack sufficient trained personnel leading to misdiagnosis or slow diagnosis. Microscopy is also time consuming and it needs laboratory infrastructure, good electricity and good maintenance of equipments which might not be easily obtained in the remote or resource limited location (O'Meara et al., 2011).

### **2.3.2 Rapid Diagnostic Tests (RDTs)**

Rapid Diagnostic Tests (RDTs) are immunochromatographic tests that are meant to identify certain malaria antigens in the blood of a patient. The common antigens of these tests include histidine-rich protein 2 (HRP2) or parasite lactate dehydrogenase (pLDH). RDTs are portable and give results within 15-20 minutes without having to use the specialized laboratory apparatus (Murray et al., 2008).

Clearly, the main benefit of RDTs is their simplicity and speed. They can be used by the least trained healthcare providers and are especially useful in the rural or peripheral health facilities where microscopy facilities are never available. RDTs have played a large role

in diagnostics of malaria in most of these endemic areas and have helped reduce the practice of presumptive treatment (Bell et al., 2016).

Nevertheless, RDTs possess limitations as well, which impact the reliability in their diagnostic tests. Low parasite density results in a reduction in their sensitivity, which generates false-negative results. Moreover, not all the *Plasmodium falciparum* strains contain the HRP2 gene, which makes it ineffective in some localities to use HRP2-based RDTs. RDTs do not also give a density or differentiation of species other than the selected antigens, which restricts their application in end-to-end case management of malaria (Cheng et al., 2014).

### **Comparison of Microscopy and RDTs**

The difference between microscopy and RDTs is very high regarding their efficiency, accuracy and operational requirements. Microscopy has greater diagnostic quality, and species detection, though requiring personnel with high skills, a laboratory facility and more processing time. Conversely, RDTs have quick and easy to use diagnosis, low training, limited diagnostic capabilities, and can be reduced in sensitivity under some conditions. Although microscopy is the reference standard, RDTs have become a valuable complementary instrument especially in places where there are limited laboratory resources (McMorrow et al., 2011).

The shortcomings of microscopy and RDTs highlight the need to have better diagnostic methods, which are more precise, efficient, and scalable. These issues have spurred the development of new diagnostic solutions, such as Artificial Intelligence-powered

systems, that seek to improve the diagnostic performance without relying on the medical knowledge and the infrastructural limitations of the human factors.

## **2.4 Artificial Intelligence as a Diagnostic Technique**

Artificial Intelligence (AI) is the capacity of computer systems to conduct functions traditionally seen as intelligence tasks (i.e., learning, reasoning, pattern recognition, and decision-making). AI has acquired a lot of popularity in the healthcare industry in the past few years because it can help increase the accuracy of the diagnosis and improve its efficiency as well as aid with the clinical decisions. The AI systems can discover complex trends in huge amounts of medical data that human experts might struggle to identify and that might even take time (Topol, 2019).

One of the main elements of AI in medical diagnostics is the use of a technique called Machine Learning (ML), where the system acquires a method of learning by analyzing data and becoming more effective as time passes without being explicitly programmed. These machine learning models can be trained based on past data and can then be used to predict or classify new data. Medical ML has found application in healthcare to solve medical image, disease identification, risk detection, and treatment prescription (Esteva et al., 2019).

There are three broad types of machine learning methods, which are supervised learning, unsupervised learning, and reinforcement learning. Training models to perform classification or prediction tasks using labeled datasets is called supervised learning and is frequently used in the diagnosis of diseases. Unsupervised learning aims to find the hidden patterns or structures of unlabeled data and it is frequently applied in clustering

and anomaly detection. Reinforcement learning is the interaction-based learning that is developed with regard to the environment and has new applications in treatment optimization and decision support systems (Jordan and Mitchell, 2015).

There are a number of benefits associated with the use of AI and machine learning in healthcare diagnostics. By using AI-based diagnostic systems, the large data can be processed quickly, human error can be reduced, and results can be consistent regardless of fatigue or subjectivity. AI models have in medical imaging performed equally to or even better than human experts. Such abilities render AI especially appropriate in terms of diagnostic work in resources-constrained settings, where the number of special healthcare workers is low (Litjens et al., 2017).

Although it has these benefits, there are significant issues with the application of AI as a diagnostic method as well. The quality and representativeness of training data is very crucial to the performance of AI models, and inaccurate prediction can happen when using biased or incomplete training data. Moreover, several AI models are black boxes and, therefore, their answers cannot be understood and relied upon by clinicians. Ethical issues, data privacy, and the necessity to obtain regulatory approval also make the use of AI-based diagnostic systems in healthcare facilities widely spread (Shen et al., 2017).

However, the current developments in computational ability, data access, and algorithm development still reinforce the justification of AI in medical diagnostics. Since medical institutions are becoming more demanding of scalable and reliable diagnostic solutions, AI-based solutions are being investigated as one of the complementary solutions to the established diagnostic methods. These innovations are a solid basis on the use of

Artificial Intelligence in malaria screening and diagnosis that is further studied in the following section.

## **2.5 Related Works**

More recent literature has been done that has dwelled more on how Artificial Intelligence techniques can be put into use to enhance malaria screening and diagnosis, especially as a response to the shortcomings of the more traditional diagnostic processes. To a large extent, the use of machine learning and deep learning models to automate the process of detecting parasites, increase the accuracy of the diagnostic process, and decrease the reliance on human expertise is used in these studies.

Rajaraman et al. (2020) suggested a malaria detection system based on deep learning along with convolutional neural networks that are trained on microscopic blood smear images. The experiment has prioritized the application of the method of data augmentation to enhance the model robustness to the variation in staining and image quality. The experimental outcomes showed a high diagnostic accuracy but the model needed high-quality labeled datasets and extensive computing resources, which meant that it could not be practically applied in low-resource health care environments.

Yang et al. (2020) designed an automated malaria diagnosis system with CNN structures that are optimized to identify parasites. The model was found to be very sensitive and specific on publicly available datasets. Although the system performed well, the authors indicated that the system had minimal validation in the real-world clinical setting, which casts doubt on its generalizability in real-life deployment environments.

Al-Doghman and Al-Fatlawi (2020) investigated the performance of deep learning models in the diagnosis of malaria and compared them to the traditional machine learning methods. They found out that deep learning methods were much stronger and more successful than traditional classifiers because of their higher accuracy and resilience. Nevertheless, the paper has identified the fact that deep learning models are not explainable as a significant obstacle to clinical adoption.

Osei et al. (2020) suggested an ensemble machine learning method to detect malaria parasites based on the microscopic image. The system enhanced the overall diagnostic accuracy and minimized the false positives by using several classifiers. However, the added complexity of the models also meant that the cost of computing went up, something that could impact the implementation on low-end devices that are regular in rural healthcare centers.

Abdar et al. (2021) provided a thorough review of machine learning implementation in the diagnosis of infectious diseases, malaria being one of them. The analysis of the studies reviewed the most popular algorithms, datasets, and metrics of evaluation, highlighting the significance of model interpretability and validation. The authors found that there is a gap in the progression of integrated diagnostic models that can be applied in a real-life clinical setting.

The computerized system proposed by Rajaraman et al. (2021) is a mobile-oriented deep learning platform that is capable of facilitating the process of screening malaria in the areas with scarce resources. The framework was based on lightweight CNN architecture so that it can be deployed on mobile devices. Although the accuracy of the system was

promising, the issues connected with the processing speed and compatibility of the devices were observed.

Kumar et al. (2021) came up with the hybrid model of malaria parasite diagnosis based on the combination of convolutional neural networks with support vector machines. The hybrid method enhanced the performance of classification by making use of deep feature extraction and conventional classification methods. Nevertheless, the system complexity of the model was added by the fact that massive preprocessing is required.

Liang et al. (2022) tested several varieties of deep learning models to diagnose malaria and how the bias of the data set affects the results of the modeling process. The researchers found out that the models that were trained on small data sets tended to perform poorly when they were tested with data that was not seen. To enhance the reliability, the authors suggested standard datasets and interregional validation.

Nkiruka et al. (2022) examined the implementation of AI-based malaria diagnostic tools in the rural healthcare environments. They have shown that AI-facilitated diagnosis enhanced the efficiency of screening and minimized the workload of healthcare professionals. However, large-scaled implementation was limited by infrastructural difficulties, including unstable power supply and data accessibility.

Sharma et al. (2023) created a lightweight convolutional neural network that is optimized to achieve diagnosis of malaria on malaria specimen-driven low-power computing platforms. In the proposed model, the diagnostic accuracy was reasonable and the number



of computations decreased substantially. Although these advantages were made, the research did note lower performance when compared to bigger models.

The proposal of Ahmed et al. (2023) is a deep learning-based malaria screening system based on transfer learning. The system minimized the training time and enhanced the classification rate by using pre-trained models. Nevertheless, the method still needed a large amount of labeled data to perform fine-tuning.

Mensah et al. (2024) discussed the possibility of integrating AI-based malaria diagnosis into the current clinical decision support systems. Their structure showed better diagnostic consistency and decision-making. Integration complexity and interoperability issues were the limitations that were mentioned in the study.

Zhang et al. (2024) presented a multi-class deep learning model that can detect various species of Plasmodium using a blood smear image. The system enhanced species-level diagnosis but needed high-resolution images, which could not be used in simple healthcare premises.

Okafor et al. (2025) created an artificial intelligence-based malaria screening system that integrates image-based and clinical data. This combined system increased the reliability of diagnostic systems as opposed to those based only on images. Nevertheless, data fusion complicated systems and data management needs.

Adeyemi et al. (2025) designed a cloud-based AI malaria patient diagnosis system to facilitate remote healthcare facilities. This system was very accurate and scalable yet it

was very dependent on steady internet connection which is not always available in endemic areas.

Okorie et al. (2025) suggested a malaria screening and diagnosis system based on AI tools through deep learning models to microscopic blood smear images. The system used image preprocessing, automated parasites detection and classification of results to aid in clinical decision-making. High diagnostic accuracy and decreased time of analysis were demonstrated by experimental analysis over manual microscopy. The study, however, mentioned challenges with model interpretability as well as the necessity of large-scale validation with heterogeneous datasets prior to large-scale clinical implementation.

On the whole, these works show that AI-malaria screening and diagnosis systems have developed significantly. Nevertheless, issues of interpretability of model, data quality, limitations on infrastructure, and systems integration remain. These gaps affect the emergence of a decent and scalable Artificial Intelligence-powered malaria screening and diagnosis support system, which is the focus of the present study.

**2.6 Summary of Related Works**

| S/N | Aurthor | Title | Summary | Limitations |
|-----|---------|-------|---------|-------------|
|-----|---------|-------|---------|-------------|

|   |                                 |                                                                 |                                                                                                                                                            |                                                               |
|---|---------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1 | Rajaraman et al. (2020)         | Deep-Learning Based Malaria detection Using Microscopic Images  | Put forward an automatic malaria parasite in blood smear images on a CNN-based system that included data augmentation to achieve high diagnostic accuracy. | Large labeled sets and high computing resources are required. |
| 2 | Yang et al. (2020)              | Automated Malaria Diagnosis Using Convolutional Neural Networks | Designed a CNN model that was sensitive and specific with regard to the diagnosis of malaria on publicly available datasets.                               | Minimal confirmation based on actual clinical data.           |
| 3 | Al- Doghman & Al-Fatlawi (2020) | Deep learning Techniques for Malaria Diagnosis                  | Comparisons between deep learning models and the traditional ML methods and the results indicated better performance in deep learning methods.             | Absence of model explainability to use in clinical practice.  |
| 4 | Osei et al. (2020)              | Ensemble Learning for Malaria Parasite Detection                | Applied machine learning methods to ensemble learning to enhance the accuracy of malaria parasite classification.                                          | More complicated computations and time.                       |
| 5 | Abdar et al. (2021)             | Machine learning for Infectious Disease Diagnosis               | Surveyed ML applications in the diagnosis of infectious diseases, with a                                                                                   | Failed to present a workable implementation structure.        |

|    |                         |                                                          |                                                                                                                             |                                                                  |
|----|-------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|    |                         |                                                          | focus on evaluation metrics and interpretability issues.                                                                    |                                                                  |
| 6  | Rajaraman et al. (2021) | Mobile-Based Deep Learning System for Malaria Screening  | Created a light convoluted neural net model to screen malaria on a mobile platform that fits in a low resource environment. | Limitation of compatibility of devices and speed of processing.  |
| 7  | Kumar et al. (2021)     | Hybrid CNN-SVM Model for Malaria Detection               | CNN feature extraction with SVM classification to enhance diagnostic accuracy.                                              | Needs a lot of pre-processing and more complexity in the system. |
| 8  | Liang et al. (2022)     | Evaluation of Deep Learning Models for Malaria Diagnosis | Evaluated various CNN models and revealed the impact of dataset bias on model performance.                                  | Weak extrapolation on invisible data sets.                       |
| 9  | Nkiruka et al. (2022)   | AI-Based Malaria Diagnosis in Rural Healthcare           | Researched how AI tools affect the efficacy of malaria diagnosis in rural areas.                                            | Poor infrastructure and irregularity of data.                    |
| 10 | Sharma et al. (2023)    | Lightweight CNN for Malaria Detection                    | A low-power CNN model with an acceptable accuracy has been proposed with lower computation cost.                            | Poorer accuracy than bigger deep learning models.                |
| 11 | Ahmed et al. (2023)     | Transfer Learning for Malaria Screening                  | Transfer learning to enhance accuracy of malaria                                                                            | Continued reliance on huge labelled datasets.                    |

|    |                       |                                                             |                                                                                                                      |                                                                     |
|----|-----------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|    |                       |                                                             | diagnosis and minimise time taken during training.                                                                   |                                                                     |
| 12 | Mensah et al. (2024)  | AI-Integrated Clinical Decision Support for Malaria         | Introduce AI into clinical decision support systems to promote a higher degree of consistency in malaria diagnosis.  | System integration problems and interoperability.                   |
| 13 | Zhang et al. (2024)   | Multi-Class Deep Learning Model for Plasmodium Detection    | Trained a deep learning system that was able to recognize several species of Plasmodium.                             | Needs optimal performance of high-resolution images.                |
| 14 | Okafor et al. (2025)  | AI-Driven Malaria Screening Usage Image and Clinical Data   | Image analysis with clinical data to improve diagnostic reliability on malaria.                                      | Additional complexity of the system and data handling requirements. |
| 15 | Adeyemi et al. (2025) | Cloud-Based AI Malaria Diagnostic System                    | Presentation of a cloud-based AI system to diagnose malaria in remote medical clinics on a larger scale.             | Relying on internet connectivity.                                   |
| 16 | Okorie et al. (2025)  | Deep Learning Framework for Malaria Screening and Diagnosis | Developed an artificial intelligence framework with preprocessing, detection and classification to aid in diagnosis. | Low model interpretability and wider validation.                    |

## CHAPTER THREE

### 3.1 System Analysis and Design Strategy

This research will employ a system analysis and design-based approach in developing and testing a malaria screening system, which will be artificial intelligence-based. The methodology selected is based on system modeling, process definition, performance analysis as opposed to clinical deployment, which is appropriate in undergraduate research study where the aim is to solve the problem using modern information technology.

The system analysis step is a thorough analysis of the current malaria diagnostic process where major issues like relying on expert microscopists, manual analysis thus time consuming and variability in diagnostic accuracy have been identified. The challenges inform functional and non-functional requirements of the proposed system and make sure that the designed solution directly responds to identified gaps in efficiency, reliability and accessibility.

This is a modular system architecture design strategy, with every element in the proposed system having a specific purpose in the entire workflow. These functionality pieces are data acquisition, data preprocessing, model training and testing as well as performance evaluation. Modularity increases the understandability of systems, their maintainability, and scalability, and enables individual analysis and refinement of stages of the artificial intelligence pipeline.

Unified Modeling Language (UML) is the main modeling tool used to have a clear representation of system functionality and interactions. UML diagrams visually describe system behaviour, structure and flow of operation. In particular, the use case diagrams will

depict the interaction between the user and the system, the class diagram will show the structural relationship among the components of the system, and the flowcharts will show the chronological steps of processing the malaria screening and diagnosis. Such visual models give a standard and easy to decipher visualization of the system to both technical and non technical stakeholders.

The strategy implementation is executed in a simulative setting utilizing publicly available malaria data and standard machine learning packages. This guarantees the reproducibility of the results and also does away with the issues of ethical considerations in direct human or clinical experimentation. The system has predetermined evaluation metrics that are used to test the system and objectively evaluate the diagnostic performance of the system.

On balance, the selected system analysis and design approach would make the solution suggested adequate, comprehensive, and consistent with the current IT problem-solving traditions, but it is realistic in scale and implementation.

### **3.2 UML Diagrams**

In this research, Unified Modeling Language (UML) diagrams are employed to have a clear and well-organized representation of the proposed malaria screening system based on AI. UML diagrams assist in graphic representation of the manner the users interrelate with the system, the arrangement of system parts, and the movement of information within the

system as it functions. UML is useful in enhancing clarity in a system, facilitating appropriate system design and making the solution proposed simple to adopt and understand.

The UML diagrams that were employed in this research are Use Case Diagram, Class Diagram, and System Flowchart.

### **3.2.1 Use Case Diagram**

The use case diagram is used to depict the communication between external users (actors) and the suggested malaria screening system. It specifies the functional requirements of the system by demonstrating what actions can be done and the response of the system to the actions.

The central actor of the proposed system is the Medical Personnel, as he/she will be involved with the system to post blood smear images and see the diagnostic results. The System Administrator who is the secondary actor is concerned with system maintenance and dataset management. The data uploaded is processed by the system itself and diagnostic outputs generated.

The most significant applications of the system are:

- Upload blood smear image



- Preprocess image data
- Perform malaria screening
- Display diagnosis result
- Evaluate model performance

As Figure 3.1 indicates, the user uses the system by uploading an image of a blood smear, which will trigger the malaria detection process and see the classification results. The following diagram shows the functional requirements of the system as per the user perspective.

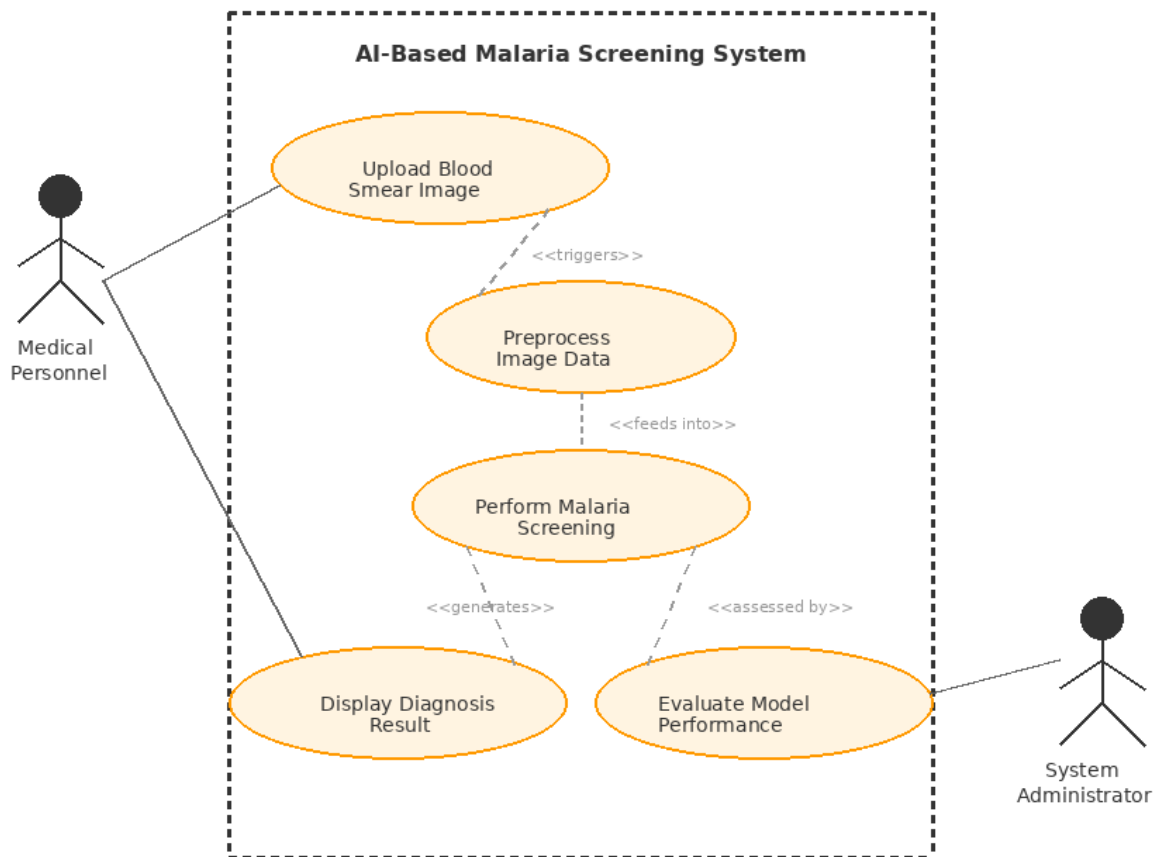


Figure 3.1: Use Case Diagram of the Proposed AI-Based Malaria Screening System

### 3.2.2 Class Diagram

The class diagram indicates the structural design of the proposed system in which the fundamentals containing the key classes, their attributes, and the methods are discussed as well as the connections between these fundamental units. It gives a roadmap on the way the system components are arranged and also the way they communicate with each other.

The main classes that will be observed in the proposed system are:

- Dataset: Image storage and retrieval.
- Preprocessor: Normalizes image, sizes images and removes noise.
- MalariaDetectionModel: Trains and tests AI model.
- EvaluationModule: Calculates performance indicators.
- UserInterface: Interacts with users of the system.

These classes have relationships with each other and some of the relationships include association and dependency which guarantees a smooth flow of data between input and output. The class diagram facilitates modularity and the system becomes easier to maintain and extend.

Important classes in the class diagram included in Figure 3.2 are Dataset, Preprocessor, MalariaDetectionModel, EvaluationModule and UserInterfaces. All these categories describe data movement in the system, including data acquisition, preprocessing, model training, prediction, and performance analysis.

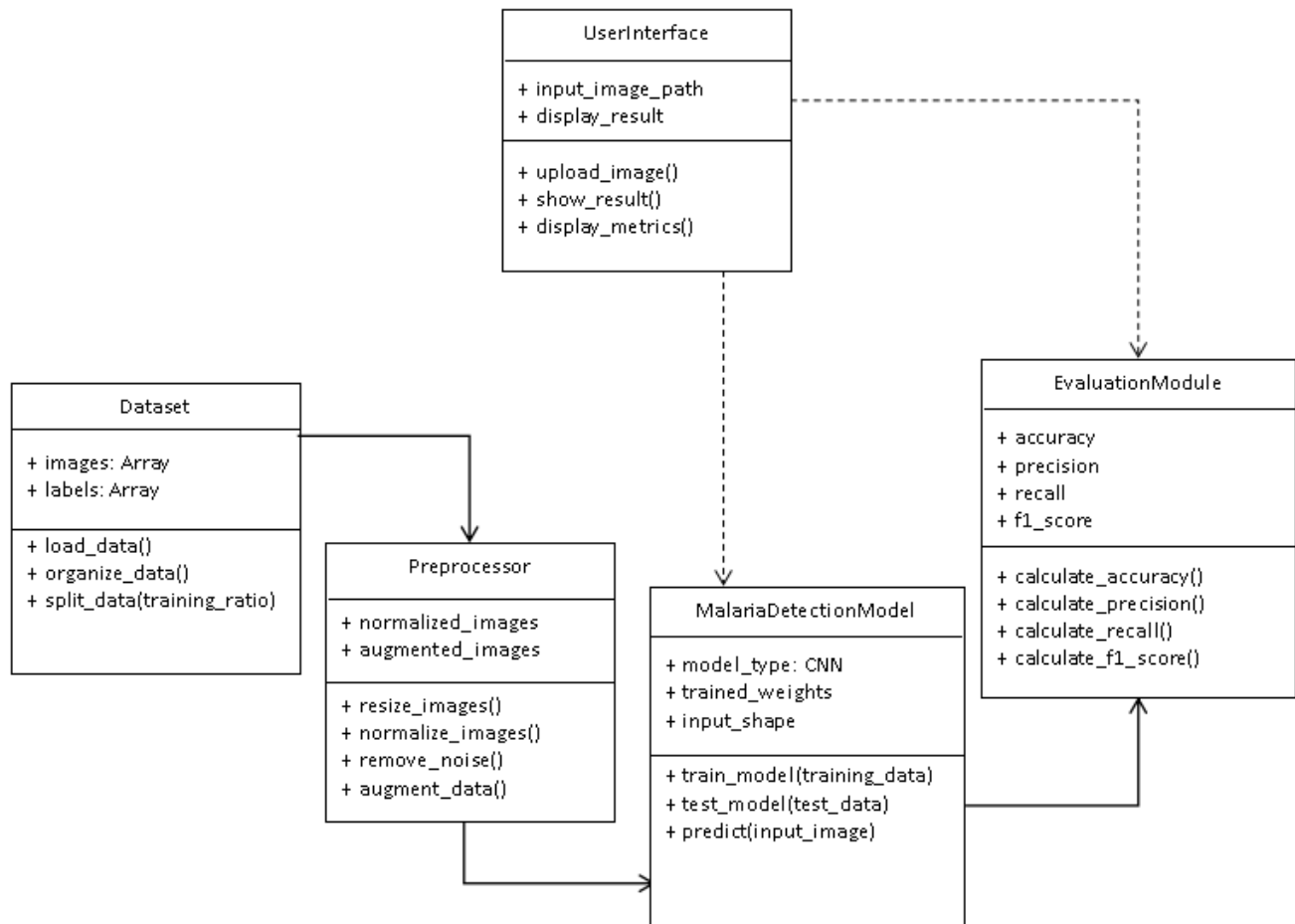


Figure 3.2: The Proposed AI-Based Malaria Screening System Class Diagram.

### 3.2.3 System Flowchart

The system flow chart shows the sequential flow of actions that are entailed in the malaria screening process. It demonstrates the flow of data across the system starting at the input point to the completion of the output.

Flow starts by taking pictures, then goes through data preprocessing, feature extraction, model training, and testing and eventually generation of results. To find out whether parasitized malaria parasites are present or not, decision points are added. The flowchart gives the clear idea of the system logic and the steps of operations.

The flowchart in Figure 3.3 illustrates a step-by-step process which begins with the image acquisition, preprocessing, training and testing the model, and finally with an output of the result of the malaria detection. This flow chart gives a real insight on how the proposed system will be operational.

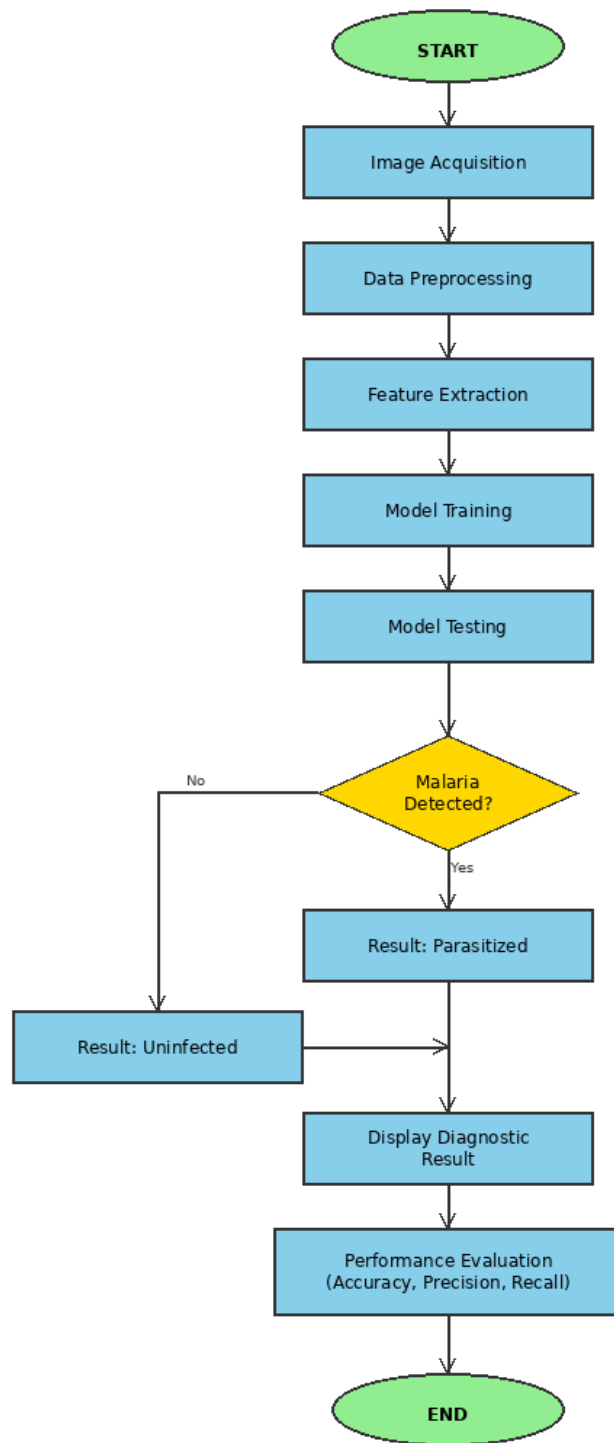


Figure 3.3: Flowchart of the Proposed AI-Based Malaria Screening System

### **3.3 Overview of the System**

The suggested system is an AI-based malaria screening system aimed to aid in the automated detection of the malaria parasites in the microscopic blood smear images. The system works within a simulated environment and combines data processing, machine learning, and evaluation in order to enable correct and efficient malaria diagnosis.

On a high-level, the system starts with the acquisition of blood smear images that are communicated by the medical personnel by the system interface. These images are the main input to the system and they are stored in the dataset module where they are then processed. When the images have been uploaded, the preprocessing module runs critical procedures like image re-size, image normalization, and noise removal in order to improve the quality of an image and to provide uniform results in case of model training and testing.

After processing the images, the processed images are fed to the malaria detecting model. This element deals with the learning patterns of malaria infected and non-infected blood samples with the help of machine learning methods. A subset of the data is used to train the model and is then tested on unseen data to determine its capability to generalize and make correct predictions.

The system has diagnostic outputs that reveal the presence of malaria parasites in the blood smear images after the training and testing phases. The system interface provides the user

with these results in a clear and interpretable format. Moreover, the performance metrics including accuracy, precision, recall, and F1-score are also calculated by the evaluation module, in order to objectively determine the effectiveness of the proposed system.

The general working process of the system is directed by the UML diagrams and flowchart in Section 3.2. The use case diagram establishes user interaction, the class diagram establishes the internal system structure and the flowchart establishes the sequential running of the system processes. All these models will guarantee that there is a coherent and well-structured system design which fits the purpose of this study.

### **3.4 Data Collection**

This study is ethical in terms of data collection, as the publicly available malaria datasets were used, thus making it ethical, reproducible, and appropriate to conduct research in a simulated implementation environment. The datasets are microscopic images of blood smear that have been either labeled in malaria infection or not, which are common in the research of machine learning models in malaria diagnostics.

The images of blood smears are images of thin films of blood taken under the normal laboratory microscopes. These pictures have red blood cells some of which can be infected with plasmodium parasites. The tagged characteristics of the datasets can facilitate the supervised learning by enabling the proposed system to acquire distinguishing characteristics between infected and uninfected samples.



The data sources in this study are based on credible open-access resources which have been frequently used in the malaria detection studies. These datasets are in the form of an image like JPEG or PNG, and the datasets are arranged into predefined categories based on the diagnosis labels. It includes only images in a clean visual state and valid labels to be reliable in the process of models training and testing.

In the course of this study, no direct information about patients or the healthcare institutions is gathered. Therefore, no physical contact with the human subjects or personal information that is sensitive about the patient is present. This methodology eradicates the ethical issues concerning privacy and consent and enables the research to be devoted to the system design and performance analysis.

Overall, the chosen data collection methodology helps to achieve the development of the proposed system with the help of reliable and well-structured data without violating ethical standards and the validity of the experiment.

### **3.5 Data Preprocessing**

Preprocessing data is a very important step in the suggested system because the quality of input data has a great impact on the performance of machine learning models. In this paper, the blood smears images recorded are preprocessed to enhance the quality of the images, minimize noise and consistency throughout the dataset prior to training and testing of the model.

Image resizing and normalization is the initial stage of the preprocessing. All pictures are also downsized to equal size so that they can fit the machine learning model and to minimize computation complexity. Positions of pixels are normalized to a standard range and this is helpful to stabilize the learning process and also enhances model convergence when training.

This is followed by noise reduction whereby the undesirable artifacts that could be present in the microscopic images are removed. These artifacts may be the result of staining differences, light differences, or differences in the settings of the microscope. Simple filtering processes are employed to advance significant visual characteristics and to restrain the background noise that is unnecessary.

The use of data augmentation methods is used to enhance the robustness of the model and minimize the possibility of overfitting. These methods are artificial expansion and diversification of the data, which is made through rotating, flipping, and scaling existing images. By increasing data augmentation, the model will learn the features that are not

dependent and be more effective when new samples that have not been seen previously are presented.

Lastly, the preprocessed images will be sorted into the training and also the testing set so that the information will be prepared to be used in the next step of correlation, training of the model, and also the performance evaluation.

### **3.6 Correlation, Training and Testing Models.**

Correlation analysis is conducted in this research to perceive the connection between the features of an input data and the assigned labels of diagnostic features. This will assist in making sure that the features that are extracted in the picture have a bearing on the classification of the blood smear pictures as infected with malaria or not. Through these relationships, irrelevant or low contribution features can be reduced thus enhancing model learning efficiency.

The training process would entail exposing the proposed malaria detection model to a subset of the preprocessed data where the model would be learning the distinguishing patterns of malaria parasites. A supervised learning approach is also used because the dataset consists of labeled elements. The model is also used in the process of training where its internal parameters are adjusted iteratively to reduce classification errors and maximize prediction accuracy.

The dataset is split into a training and a testing set to determine the generalization capability of the model. The model is built using the training set, and the testing set is comprised of hidden data which is used to evaluate the performance of the model. This division provides that the results of the evaluation are a true indication of the capacity of the system in diagnosing new samples and not just the training information.

Testing involves the implementation of the model that has undergone training on the testing data and evaluation of the results that the model predicts against the real labels. The results of this stage in terms of performance allow seeing how well and how reliable the suggested system is and serve as the foundation of further evaluation based on the predefined metrics.

### **3.7 Study Population**

The research study population will be the blood smear image sample applied in the simulated test of the proposed system. These samples are red blood cells that are derived from patients in malaria infected areas, which are largely used in studies on malaria diagnosis. The population of the study is also indirect because the review will be based on secondary data and no interaction with human subjects will be carried out.

The blood smear images involved in undertaking the present study are those that were annotated as malaria infected and non-infected. They are labels grounded on previous expert verification and used as ground truth during training and evaluation of the suggested

machine learning model. The data is not related to any demographic information including age, gender, or medical history that guarantees anonymity and privacy.

Geographically, the statistics indicate malaria-infested environments, especially those areas where malaria continues to pose a significant challenge to the general population. Nevertheless, the research does not target a particular nation or health care facility since the objective is to assess the technical functionality of the suggested system and not to carry out an epidemiological research based on the population.

The research will have no direct interaction with patients or healthcare providers because the study will use publicly available datasets, and no ethical permission is needed. This method will guarantee adherence to ethical research principles and will enable a successful analysis of the offered system of AI-based malaria screening.

### **3.8 Data Analysis Technique**

The data analysis methodology that will be used in this paper is the use of supervised machine learning techniques to label the blood smear images either as infected with malaria or not. The system performs the analysis of the preprocessed images by reading out the meaningful features through the learning process and then the features are used to make predictions using patterns in the training data.

The main machine learning model employed is a Convolutional Neural Network (CNN) since this type of model has been successfully applied in tasks related to the classification of images. The CNN is now able to automatically learn hierarchical features like the morphology of parasites, cell shapes and staining patterns without manually having to engineer features. This will enable the system to detect minute visual clues that differentiate infected cells and healthy cells.

The training dataset is iteratively trained on the model, in which the model is modified to reduce the variance between predicted and actual labels. The model is tested on the testing data after it has been trained to see whether it was able to predict well. The predictions can be compared against the ground truth labels in order to determine the accuracy, reliability and generalizability of the system.

This is done to make the analysis objective, reproducible, and quantitative to give a clear measure of the capability of the system to undertake automated malaria screening. The obtained results of this method serve as the foundation of the performance assessment in terms of conventional indicators like the accuracy, precision, recall and F1-score.

### **3.9 Evaluation Metrics**

In order to be objective in the evaluation of the performance of the proposed AI-based malaria screening system, there are a number of standard evaluation measures which are

used. These measures are measures of diagnostic accuracy and capability of the system to detect the presence of malaria and non-malaria blood smear images in the correct way.

### 1. Accuracy

Accuracy is a measure of the percentage of samples that are correctly classified of the total number of samples examined. It gives a broad picture on how the system is performing as a whole.

### 2. Precision

Precision is the fraction of the number of positive cases that are actually predicted that is positive compared to the number of cases that are predicted to be positive. High accuracy means that the system creates less false positives, which is essential to prevent avoidable treatment.

### 3. Recall (Sensitivity)

The recall is the rate of correctly identified cases of actual positive cases by the system. The high recall rate will make sure that no cases of the malaria infection are missed because they are correctly identified.

### 4. F1-Score

F1-score, the harmonic average of the recall and the precision, gives only one number where the false negative and false positives can be balanced. It comes in handy especially in the case of an imbalanced dataset as in the case of medical image classification.

These metrics of evaluation are computed using model prediction on the testing data set, i.e. unknown blood smear images. Using these metrics, the study will be able to establish the effectiveness, reliability and robustness of the proposed system in carrying out automated malaria diagnosis.



## REFERENCES

- Abdar, M., Agarwal, S., & Sharma, S. (2021). Artificial intelligence in healthcare: A review on machine learning techniques and applications. *Journal of Healthcare Engineering*, 2021, 1–20. <https://doi.org/10.1155/2021/6614006>
- Adeyemi, O., Balogun, A., & Adebayo, S. (2025). A cloud-based artificial intelligence framework for malaria diagnosis in remote healthcare facilities. *Journal of Medical Systems*, 49(2), 1–14.
- Ahmed, K., Hassan, R., & Ali, M. (2023). Transfer learning approaches for automated malaria screening using deep convolutional neural networks. *Biomedical Signal Processing and Control*, 82, 104566.
- Al-Doghman, F., & Al-Fatlawi, A. (2020). Deep learning techniques for malaria diagnosis: A comparative study. *Computers in Biology and Medicine*, 123, 103903. <https://doi.org/10.1016/j.combiomed.2020.103903>
- Bell, D., Wongsrichanalai, C., & Barnwell, J. W. (2016). Ensuring quality and access for malaria diagnosis: How can it be achieved? *Nature Reviews Microbiology*, 14(7), 448–459.
- Bousema, T., Okell, L., Felger, I., & Drakeley, C. (2018). Asymptomatic malaria infections: Detectability, transmissibility and public health relevance. *Nature Reviews Microbiology*, 12(12), 833–840.
- Cheng, Q., Gatton, M. L., Barnwell, J., & Chiodini, P. (2014). *Plasmodium falciparum* parasites lacking histidine-rich protein 2 and 3: A review of the worldwide evidence. *Malaria Journal*, 13, 123.

Esteva, A., Robicquet, A., Ramsundar, B., et al. (2019). A guide to deep learning in healthcare. *Nature Medicine*, 25(1), 24–29.

Gething, P. W., Casey, D. C., Weiss, D. J., et al. (2017). Mapping *Plasmodium falciparum* mortality in Africa between 1990 and 2015. *Nature*, 526(7572), 207–211.

Jordan, M. I., & Mitchell, T. M. (2015). Machine learning: Trends, perspectives, and prospects. *Science*, 349(6245), 255–260.

Kumar, S., Verma, B., & Singh, A. (2021). Hybrid CNN–SVM based malaria parasite detection using microscopic blood smear images. *Applied Soft Computing*, 109, 107488.

Landier, J., Parker, D. M., Thu, A. M., et al. (2016). Effect of early diagnosis and treatment on malaria transmission. *Malaria Journal*, 15, 1–12.

Liang, Z., Powell, A., Ersoy, I., et al. (2022). CNN-based image analysis for malaria diagnosis: Dataset bias and generalization issues. *IEEE Journal of Biomedical and Health Informatics*, 26(3), 1234–1245.

Litjens, G., Kooi, T., Bejnordi, B. E., et al. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88.

McMorrow, M. L., Aidoo, M., & Kachur, S. P. (2011). Malaria rapid diagnostic tests in elimination settings. *The American Journal of Tropical Medicine and Hygiene*, 84(5), 695–702.

Mensah, E., Owusu, A., & Boateng, K. (2024). Integrating artificial intelligence into clinical decision support systems for malaria diagnosis. *Health Informatics Journal*, 30(1), 1–15.

- Moody, A. (2020). Rapid diagnostic tests for malaria parasites. *Clinical Microbiology Reviews*, 15(1), 66–78.
- Murray, C. K., Gasser, R. A., Magill, A. J., & Miller, R. S. (2008). Update on rapid diagnostic testing for malaria. *Clinical Microbiology Reviews*, 21(1), 97–110.
- Nkiruka, O., Eze, P., & Okorie, C. (2022). Artificial intelligence–based malaria diagnosis in rural healthcare environments. *African Journal of Medical Informatics*, 9(2), 45–56.
- Ochola, L. B., Vounatsou, P., Smith, T., & Mabaso, M. L. H. (2016). The reliability of diagnostic techniques in malaria case management. *Malaria Journal*, 15, 1–9.
- Okafor, C., Ibrahim, M., & Bello, A. (2025). AI-driven malaria screening using combined image and clinical data. *Artificial Intelligence in Medicine*, 138, 102528.
- Okell, L. C., Ghani, A. C., Lyons, E., & Drakeley, C. J. (2020). Submicroscopic infection in malaria-endemic populations. *Trends in Parasitology*, 25(12), 544–549.
- O’Meara, W. P., Mangeni, J. N., Steketee, R., & Greenwood, B. (2011). Changes in the burden of malaria in sub-Saharan Africa. *The Lancet Infectious Diseases*, 10(8), 545–555.
- Osei, E., Asante, M., & Boadi, E. (2020). Ensemble machine learning techniques for malaria parasite detection. *Journal of Biomedical Informatics*, 107, 103432.
- Phillips, M. A., Burrows, J. N., Manyando, C., van Huijsduijnen, R. H., Van Voorhis, W. C., & Wells, T. N. C. (2017). Malaria. *Nature Reviews Disease Primers*, 3, 17050.
- Rajaraman, S., Antani, S. K., Poostchi, M., et al. (2020). Pre-trained convolutional neural networks as feature extractors for malaria parasite detection. *PeerJ*, 8, e10123.

- Rajaraman, S., Jaeger, S., & Antani, S. (2021). Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images. *PeerJ*, 9, e11399.
- Sharma, A., Gupta, R., & Verma, S. (2023). Lightweight deep learning models for malaria detection on low-power devices. *Computers in Biology and Medicine*, 155, 106590.
- Tangpukdee, N., Duangdee, C., Wilairatana, P., & Krudsood, S. (2009). Malaria diagnosis: A brief review. *The Korean Journal of Parasitology*, 47(2), 93–102.
- Tizifa, T. A., Kabaghe, A. N., McCann, R. S., et al. (2018). Prevention efforts for malaria. *Malaria Journal*, 17, 1–10.
- Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56.
- White, N. J., Pukrittayakamee, S., Hien, T. T., et al. (2018). Malaria. *The Lancet*, 383(9918), 723–735.
- Wongsrichanalai, C., Barcus, M. J., Muth, S., Sutamihardja, A., & Wernsdorfer, W. H. (2007). A review of malaria diagnostic tools. *The American Journal of Tropical Medicine and Hygiene*, 77(6), 119–127.
- Yang, F., Poostchi, M., Yu, H., et al. (2020). Deep learning for smartphone-based malaria parasite detection. *IEEE Journal of Biomedical and Health Informatics*, 24(5), 1427–1438.
- Zhang, Y., Li, X., & Wang, J. (2024). Multi-class deep learning model for Plasmodium species identification. *Biomedical Engineering Online*, 23(1), 1–17.

